ELECTRICAL ENGINEERING
UNIT 2: MODERN CAREER TRACKS FOR ELECTRICAL ENGINEERS
Curriculum: Lovely Professional University | Department of Electrical Engineering
Reference: IEEE Spectrum Career Guide / Power Electronics Handbook by M. H. Rashid

>> 2.1 Core Power Sector & Public Sector Undertakings (PSUs)

The foundation of electrical engineering lies in national infrastructure:
- Power Generation: Thermal, nuclear, and hydro power stations (NTPC, NHPC, NPCIL, State Gencos). Focus areas include generator synchronization, exciter tuning, turbine governors, and thermodynamic heat-rate optimization.
- Power Transmission: High-voltage bulk transmission networks (Power Grid Corporation of India - PGCIL). Key disciplines: Substation automation (IEC 61850), synchrophasor PMUs, FACTS devices, and transmission line protection relays.
- Power Distribution: Discoms (Tata Power, Adani Electricity, BSES, State Discoms). Modernization tracks: Automated Metering Infrastructure (AMI), GIS mapping, distribution transformer health monitoring, and theft prevention.
- PSU Recruitment Pathways: GATE (Graduate Aptitude Test in Engineering) serves as the primary entrance gateway for Maharatna and Navratna enterprises (ONGC, IOCL, BHEL, GAIL, SAIL, PGCIL, BARC, ISRO).

>> 2.2 Power Electronics, Electric Vehicles (EV) & Motor Drives

The automotive revolution has created unprecedented demand for electrical engineers:
- EV Powertrain Architecture: On-Board Chargers (OBC), High-Voltage Traction Inverters, DC-DC Converters (400V/800V down to 12V auxiliary bus).
- Semiconductor Advancements: Transition from silicon IGBTs to Wide Bandgap (WBG) semiconductors (Silicon Carbide - SiC MOSFETs and Gallium Nitride - GaN HEMTs) achieving > 98.5% efficiency and switching frequencies > 150 kHz.
- Traction Motor Technologies: Permanent Magnet Synchronous Motors (PMSM) for high power density, Interior Permanent Magnet (IPM) for high-speed field weakening, and Brushless DC (BLDC) motors for light electric vehicles.
- Battery Management Systems (BMS): State of Charge (SoC) estimation using Extended Kalman Filters (EKF), State of Health (SoH), active and passive cell balancing, and thermal runaway prevention.

>> 2.3 Renewable Energy Engineering & Microgrids

Decarbonization is transforming the power generation architecture:
- Solar Photovoltaic Systems: Utility-scale solar farms and commercial rooftop plants. String inverters, central inverters, and Maximum Power Point Tracking (MPPT) algorithms.
- Wind Energy Systems: Doubly-Fed Induction Generators (DFIG) with partial-scale power converters, and Permanent Magnet Synchronous Generators (PMSG) with full-scale

back-to-back converters.
- Microgrid Controllers: Islanding detection (loss-of-mains protection), black-start sequencing, frequency and voltage droop control, and Battery Energy Storage Systems (BESS) integration.

>> 2.4 Industrial Automation, PLC, SCADA & Robotics

Industry 4.0 integrates electrical power actuation with digital controls:
- Programmable Logic Controllers (PLC): Structured programming based on IEC 61131-3 standard (Ladder Diagram, Function Block Diagram, Structured Text). Manufacturers: Siemens (TIA Portal / S7-1500), Rockwell Automation (Allen-Bradley ControlLogix), Schneider Electric.
- Supervisory Control and Data Acquisition (SCADA): Distributed process monitoring, alarming, historical data logging, and OPC-UA communication.
- Variable Frequency Drives (VFD): Flux Vector Control (FVC) and Direct Torque Control (DTC) for energy-efficient motor speed control across pumps, fans, conveyors, and compressors.

DETAILED WORKED EXAMPLES & ENGINEERING COMPUTATIONS

>> Example 2.1: EV Traction Motor Power and Battery Pack Sizing

Statement: An engineering design team is sizing the electric powertrain for an electric passenger vehicle with the following vehicle dynamics parameters:
- Curb mass including passengers: M = 1,600 kg
- Frontal area: A = 2.2 m^2
- Aerodynamic drag coefficient: C_d = 0.28
- Air density at 25 deg C: rho = 1.20 kg/m^3
- Coefficient of rolling resistance: C_rr = 0.012
- Acceleration due to gravity: g = 9.81 m/s^2
(a) Calculate the steady-state mechanical power (kW) required at the wheels to cruise at a highway speed of v = 108 km/h (30 m/s) on a flat road.
(b) Calculate the peak acceleration power (kW) required to accelerate from 0 to 100 km/h (0 to 27.78 m/s) in t_acc = 7.5 seconds on level ground (assuming average speed = 13.89 m/s).
(c) Determine the minimum usable battery pack capacity (kWh) to achieve a driving range of 350 km at 108 km/h, assuming powertrain efficiency eta_drive = 88% and Depth of Discharge DoD = 85%.

Step-by-Step Solution:
1. Calculation of Highway Cruising Power (at v = 30 m/s):
- Rolling resistance force: F_rr = C_rr * M * g = 0.012 * 1,600 * 9.81 = 188.35 N

- Aerodynamic drag force: F_aero = 0.5 * rho * C_d * A * v^2
F_aero = 0.5 * 1.20 * 0.28 * 2.2 * (30)^2 = 0.3696 * 900 = 332.64 N
- Total tractive force: F_total = F_rr + F_aero = 188.35 + 332.64 = 520.99 N
- Wheel tractive power: P_wheel = F_total * v = 520.99 * 30 = 15,629.7 W = 15.63 kW

2. Calculation of Peak Acceleration Power (0 to 100 km/h in 7.5 s):
- Kinetic energy imparted: Delta_E_k = 0.5 * M * (v_final)^2 = 0.5 * 1,600 * (27.78)^2 = 800 * 771.73 = 617,384 J
- Inertial acceleration power: P_acc = Delta_E_k / t_acc = 617,384 / 7.5 = 82,318 W = 82.32 kW
- Average resistance power during acceleration (at v_avg = 13.89 m/s):
F_rr_avg = 188.35 N
F_aero_avg = 0.3696 * (13.89)^2 = 71.31 N
P_res_avg = (188.35 + 71.31) * 13.89 = 3,606 W = 3.61 kW
- Peak required motor output power: P_peak = P_acc + P_res_avg = 82.32 + 3.61 = 85.93 kW (approx. 115 HP).

3. Battery Pack Capacity Sizing for 350 km Highway Range:
- Time required to travel 350 km at 108 km/h: T_drive = 350 / 108 = 3.241 hours.
- Total mechanical energy needed at wheels: E_wheel = P_wheel * T_drive = 15.63 kW * 3.241 h = 50.66 kWh
- Electrical energy drawn from battery: E_batt_usable = E_wheel / eta_drive = 50.66 / 0.88 = 57.57 kWh
- Total nominal battery capacity required (accounting for DoD = 85%):
E_nominal = E_batt_usable / DoD = 57.57 / 0.85 = 67.73 kWh.
- Specification: A standard 70 kWh Lithium-ion (NMC 811) pack perfectly satisfies the design requirement.

>> Example 2.2: Commercial Rooftop Solar PV System Design & MPPT Sizing

Statement: Design a 250 kWp grid-connected commercial rooftop solar PV plant for an engineering institute with the following technical specifications:
- Selected PV Module: Monocrystalline PERC 540 Wp
- Rated Power (P_mp) = 540 W
- Voltage at Max Power (V_mp) = 41.8 V
- Current at Max Power (I_mp) = 12.92 A
- Open-Circuit Voltage (V_oc) = 49.8 V (at 25 deg C)
- Temperature coefficient of V_oc = -0.28% / deg C
- Minimum winter cell temperature = 0 deg C; Maximum summer cell temperature = 65 deg C
- Selected Central Solar Inverter: 200 kW AC
- MPPT voltage operating range: 480 V to 850 V
- Maximum DC input voltage rating: 1,100 V
(a) Determine the minimum and maximum allowable modules per series string to stay within the inverter MPPT window.
(b) Determine the number of parallel strings and total DC peak capacity installed.

(c) Calculate the DC-to-AC overloading ratio and verify system compliance.

Step-by-Step Solution:
1. Temperature Corrected Module Voltages:
- Max Open-Circuit Voltage at 0 deg C (coldest winter day):
Delta_T_cold = 0 - 25 = -25 deg C
V_oc_cold = 49.8 * [1 + (-0.0028 * -25)] = 49.8 * [1 + 0.07] = 49.8 * 1.07 = 53.29 V
- Min Max-Power Voltage at 65 deg C (hottest summer day):
Delta_T_hot = 65 - 25 = +40 deg C
V_mp_hot = 41.8 * [1 + (-0.0035 * 40)] = 41.8 * [1 - 0.14] = 41.8 * 0.86 = 35.95 V

2. String Sizing Limits per Inverter MPPT Channel:
- Max modules per string (based on 1,100 V limit): N_max = floor(1,100 / V_oc_cold) = floor(1,100 / 53.29) = floor(20.64) = 20 modules.
- Max modules per string (to stay under 850 V MPPT max at 25 C): N_mppt_max = floor(850 / 41.8) = floor(20.33) = 20 modules.
- Min modules per string (to stay above 480 V MPPT min at 65 C): N_min = ceil(480 / V_mp_hot) = ceil(480 / 35.95) = ceil(13.35) = 14 modules.
- Optimal engineering selection: 18 modules in series per string.
- Verification: String V_oc_cold = 18 * 53.29 = 959.2 V < 1,100 V (Safe!).
- Verification: String V_mp_hot = 18 * 35.95 = 647.1 V in [480 V, 850 V] (Optimal MPPT!).

3. String and Array Sizing:
- Target array capacity = 250 kWp = 250,000 Wp
- Total modules needed = ceil(250,000 / 540) = ceil(462.96) = 468 modules.
- Number of parallel strings = 468 / 18 = 26 parallel strings.
- Total installed DC capacity: P_DC_total = 468 * 540 W = 252,720 W = 252.72 kWp.

4. DC-to-AC Overloading Ratio:
- Ratio = P_DC / P_AC_inverter = 252.72 kW / 200 kW = 1.2636 (126.4%).
- Engineering Evaluation: A DC/AC ratio of 1.25 - 1.30 is the industry gold standard, maximizing inverter capacity utilization in morning and late afternoon hours without excessive inverter clipping during midday peak.

>> Example 2.3: Industrial Variable Frequency Drive (VFD) Energy Savings

Statement: A manufacturing plant operates a 90 kW, 415 V, 50 Hz, 3-phase induction motor driving a large industrial centrifugal chilled-water pump continuously for 8,000 hours per year.
Due to seasonal production variations, the process requires only 70% of rated fluid flow (Q = 0.70 * Q_rated) for 60% of operating hours (4,800 hours), and 100% flow for the remaining 3,200 hours.
Compare the two flow modulation methods:
- Method A: Traditional mechanical throttle valve control (Power consumed at 70% flow is

86% of rated motor power).
- Method B: Electronic VFD speed control obeying Centrifugal Pump Affinity Laws (P proportional to N^3).
Assuming industrial electricity tariff of Rs. 8.50 per kWh, motor/drive efficiency of 92%, calculate:
(a) Annual energy consumption (kWh) under Method A.
(b) Annual energy consumption (kWh) under Method B.
(c) Net annual monetary savings and payback period for a VFD installation costing Rs. 5,20,000.

Step-by-Step Solution:
1. Power Consumption Calculations:
- Motor electrical input power at 100% load: P_100 = 90 kW / 0.92 = 97.83 kW.

2. Method A (Mechanical Throttling):
- At 100% flow (3,200 h): E_100 = 97.83 kW * 3,200 h = 313,056 kWh.
- At 70% flow throttled (4,800 h): Power is 86% of full load = 0.86 * 97.83 = 84.13 kW.
- Energy at 70% flow: E_70_A = 84.13 kW * 4,800 h = 403,824 kWh.
- Total Annual Energy (Method A): E_total_A = 313,056 + 403,824 = 716,880 kWh.

3. Method B (VFD Speed Control via Affinity Laws):
- Centrifugal Pump Affinity Laws state: Flow Q proportional to N; Power P proportional to N^3.
- Therefore, at 70% flow, motor mechanical power required: P_70_mech = P_rated * (0.70)^3 = 90 kW * 0.343 = 30.87 kW.
- Electrical input power via VFD (accounting for 96% VFD and 90% partial load motor efficiency = 86.4% total):
P_70_VFD = 30.87 kW / 0.864 = 35.73 kW.
- Energy at 70% flow: E_70_B = 35.73 kW * 4,800 h = 171,504 kWh.
- At 100% flow (3,200 h): Motor input with VFD (95% VFD eff): P_100_VFD = 90 / (0.92 * 0.98) = 99.82 kW.
- Energy at 100% flow: E_100_B = 99.82 kW * 3,200 h = 319,424 kWh.
- Total Annual Energy (Method B): E_total_B = 319,424 + 171,504 = 490,928 kWh.

4. Energy and Monetary Savings Analysis:
- Annual Energy Saved: Delta_E = 716,880 - 490,928 = 225,952 kWh/year.
- Annual Electricity Cost Savings: Savings = 225,952 kWh * Rs. 8.50 = Rs. 1,920,592 per year (Rs. 19.2 Lakhs/year!).
- Simple Payback Period = Capital Cost / Annual Savings = 5,20,000 / 1,920,592 = 0.27 years (approx. 3.25 months!).

>> Example 2.4: Power Quality Harmonics Analysis and Shunt Passive Filter Design

Statement: A 415 V, 50 Hz 3-phase industrial distribution panel feeds a bank of 6-pulse thyristor bridge rectifiers. A power quality analyzer records the fundamental and

harmonic currents as follows:
- Fundamental (50 Hz): I_1 = 300 A
- 5th Harmonic (250 Hz): I_5 = 60 A
- 7th Harmonic (350 Hz): I_7 = 42 A
- 11th Harmonic (550 Hz): I_11 = 25 A
- 13th Harmonic (650 Hz): I_13 = 18 A
(a) Calculate the Total Harmonic Distortion of current (THD_i) and the true RMS current drawn from the transformer.
(b) If IEEE 519 standard mandates THD_i < 8.0%, evaluate if the system violates compliance.
(c) Design a single-tuned LC passive harmonic filter tuned to the 5th harmonic (f_tune = 250 Hz) that supplies 75 kVAR of fundamental reactive power to correct the displacement power factor.

Step-by-Step Solution:
1. Calculation of Total Harmonic Current and True RMS Current:
- Harmonic sum of squares: I_harm^2 = I_5^2 + I_7^2 + I_11^2 + I_13^2
I_harm^2 = 60^2 + 42^2 + 25^2 + 18^2 = 3600 + 1764 + 625 + 324 = 6,313 A^2
I_harm = sqrt(6,313) = 79.45 A
- Total Harmonic Distortion: THD_i = (I_harm / I_1) * 100 = (79.45 / 300) * 100 = 26.48%
- True RMS Current: I_rms = sqrt(I_1^2 + I_harm^2) = sqrt(300^2 + 79.45^2) = sqrt(90,000 + 6,313) = 310.34 A.
- Compliance Evaluation: Measured THD_i = 26.48% severely violates IEEE 519 limit of 8.0%. Active or passive filtering is legally required.

2. Design of Single-Tuned 5th Harmonic Filter Branch:
- Tuning order: n = f_tune / f_1 = 250 / 50 = 5.0 (often tuned to n = 4.8 to avoid anti-resonance tolerances).
- Three-phase filter capacitive reactive power: Q_f = 75 kVAR = 75,000 VAR.
- Phase voltage: V_ph = 415 / sqrt(3) = 239.6 V.
- Fundamental capacitive reactance per phase (delta or wye):
In wye connection: X_C - X_L = V_ph^2 / (Q_f / 3) = (239.6)^2 / 25,000 = 57,408 / 25,000 = 2.296 Ohms.
- At tuning frequency omega_n = 5 * omega_1: X_Ln = X_Cn => 5 * X_L1 = X_C1 / 5 => X_L1 = X_C1 / 25.
- Net net fundamental reactance: X_net1 = X_C1 - X_L1 = X_C1 * (1 - 1/25) = (24/25) * X_C1 = 2.296 Ohms.
- Capacitive Reactance: X_C1 = 2.296 * (25 / 24) = 2.392 Ohms.
- Capacitance: C = 1 / (2 * pi * 50 * 2.392) = 1 / 751.47 = 1,330.7 microFarads per phase.
- Inductive Reactance: X_L1 = X_C1 / 25 = 2.392 / 25 = 0.0957 Ohms.
- Tuning Inductance: L = X_L1 / (2 * pi * 50) = 0.0957 / 314.16 = 0.3046 milliHenries (304.6 microH).